

MACHENG, China

WMO: 573990

Lat: 31.1309N

Lon: 114.9547E

Elev: 75

StdP: 100.42

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.7	-13.3	1.2	3.1	-10.5	1.5	3.5	7.3	6.4	6.5	4.9	1.8	0	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	36.2	26.8	35.1	26.6	34.1	26.2	28.2	33.4	27.7	32.9	27.2	32.3	3.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	22.9	30.7	26.3	22.0	30.4	25.8	21.3	30.2	91.4	33.6	88.9	32.9	86.8	32.6	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	6.7	5.8	5.0	DB	-5.5	37.9	1.6	1.0	-6.7	38.7	-7.6	39.3	-8.5	39.9	-9.7	40.6
(5)				WB	-6.8	29.0	1.4	0.7	-7.8	29.5	-8.6	29.9	-9.4	30.3	-10.4	30.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	4.4	7.0	12.0	17.8	22.7	26.2	29.3	28.7	24.7	18.9	12.5	6.5	(6)
(7)		DBStd	9.16	3.06	3.85	4.05	3.80	2.92	2.55	2.57	2.78	3.09	3.32	3.90	3.07	(7)
(8)		HDD10.0	432	175	98	24	1	0	0	0	0	0	19	114		(8)
(9)		HDD18.3	1582	431	317	199	53	3	0	0	1	33	178	367		(9)
(10)		CDD10.0	3211	2	14	87	236	394	486	597	579	439	277	93	6	(10)
(11)		CDD18.3	1320	0	0	4	38	138	236	339	321	190	52	3	0	(11)
(12)		CDH23.3	13901	0	2	32	292	1106	2322	4330	3838	1641	321	18	0	(12)
(13)		CDH26.7	6172	0	0	5	68	376	926	2213	1867	643	73	1	0	(13)
(14)	Wind	WSAvg	2.2	2.1	2.2	2.2	2.4	2.2	2.1	2.4	2.4	2.3	2.0	2.1	2.0	(14)
(15)	Precipitation	PrecAvg	1216	28	50	82	127	151	191	215	119	89	84	55	21	(15)
(16)		PrecMax	1878	60	149	176	263	352	496	848	383	233	337	155	87	(16)
(17)		PrecMin	683	0	0	10	20	33	6	22	18	5	2	0	0	(17)
(18)		PrecStd	290	16	35	37	60	74	113	167	100	59	77	41	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	21.9	26.7	31.1	33.8	35.5	37.5	37.6	35.4	31.2	25.6	18.2	(19)
(20)		MCWB	9.4	14.3	18.5	20.3	22.3	24.5	27.2	27.0	25.4	20.9	15.9	10.5		(20)
(21)		2%	DB	13.5	18.1	23.5	28.7	32.0	34.0	36.3	36.2	33.7	28.7	23.1	15.4	(21)
(22)		MCWB	7.9	11.3	15.7	19.4	21.7	24.9	27.2	27.0	24.7	19.6	15.2	9.2		(22)
(23)		5%	DB	11.5	15.7	21.3	26.7	30.5	32.7	35.3	35.1	32.0	26.9	20.8	13.4	(23)
(24)		MCWB	6.8	9.9	14.4	18.6	21.4	24.6	27.0	26.9	23.5	18.4	13.9	8.1		(24)
(25)		10%	DB	9.7	13.3	19.0	24.7	28.8	31.3	34.3	33.9	30.3	25.0	18.7	11.6	(25)
(26)		MCWB	5.8	9.0	13.3	17.4	20.8	24.3	26.9	26.5	22.8	17.7	13.0	7.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	15.6	19.9	22.9	25.1	27.7	29.0	29.0	27.2	23.7	18.6	12.2	(27)
(28)		MCDWB	14.3	20.5	25.1	28.1	30.0	32.2	33.9	33.8	32.7	28.3	22.7	15.3		(28)
(29)		2%	WB	9.1	12.8	17.3	21.2	24.0	26.8	28.3	28.2	26.4	21.5	16.6	10.4	(29)
(30)		MCDWB	12.0	15.8	21.9	26.0	28.2	31.4	33.6	33.4	31.4	25.6	20.4	13.5		(30)
(31)		5%	WB	7.6	10.9	15.5	20.0	23.1	26.1	27.8	27.6	25.6	20.2	15.3	9.1	(31)
(32)		MCDWB	10.8	14.1	19.9	24.3	27.7	30.3	33.1	32.8	30.0	24.3	18.9	12.4		(32)
(33)		10%	WB	6.2	9.5	14.0	18.7	22.2	25.5	27.4	27.1	24.6	19.1	14.0	8.0	(33)
(34)		MCDWB	8.9	12.7	17.6	22.8	26.4	29.4	32.6	32.2	28.6	23.2	18.1	10.9		(34)
(35)	Mean Daily Temperature Range	MDBR	7.5	7.8	8.6	9.1	8.7	7.6	7.1	7.7	8.5	8.8	9.0	8.3	(35)	
(36)		5% DB	MCDBR	11.6	12.4	12.4	12.2	11.7	9.8	8.8	9.3	10.7	12.3	12.6	12.3	(36)
(37)		MCWBR	6.7	6.8	6.0	5.3	4.0	3.2	2.3	2.5	3.3	4.4	5.6	6.6	(37)	
(38)		5% WB	MCDBR	10.0	9.9	10.4	9.8	8.5	7.5	8.2	8.4	8.5	9.3	9.7	10.2	(38)
(39)		MCWBR	6.4	6.1	5.8	5.2	3.8	3.3	2.7	2.8	3.5	4.3	5.1	6.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.619	0.676	0.708	0.705	0.687	0.714	0.664	0.688	0.687	0.673	0.632	0.603	(40)	
(41)		taud	1.589	1.525	1.496	1.548	1.616	1.601	1.732	1.664	1.631	1.585	1.618	1.632	(41)	
(42)		Ebn at Noon	574	592	618	648	665	645	676	651	626	586	560	558	(42)	
(43)		Edh at Noon	223	260	285	280	263	266	232	246	245	240	214	204	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.11	2.40	3.50	4.18	4.46	4.41	4.83	4.48	3.85	3.11	2.52	2.10	(44)	
(45)		RadStd	0.33	0.59	0.28	0.44	0.42	0.43	0.54	0.51	0.46	0.43	0.33	0.38	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page